Machine Learning

CSE 4621

Presented by

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Assistant Professor

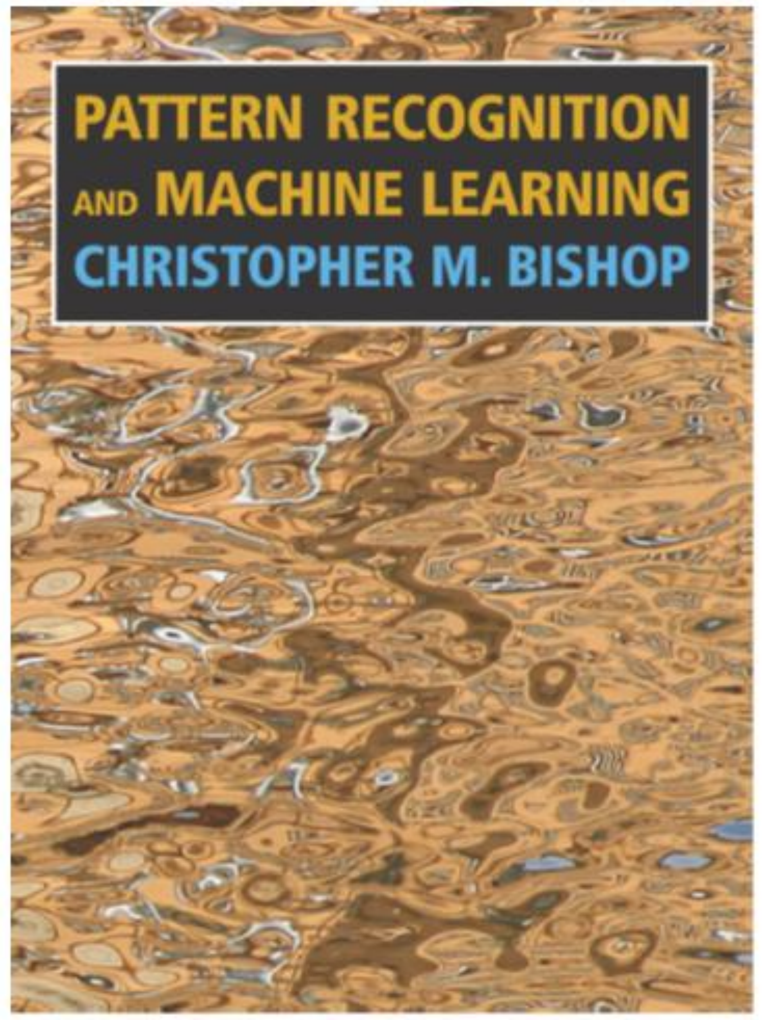
Course objective

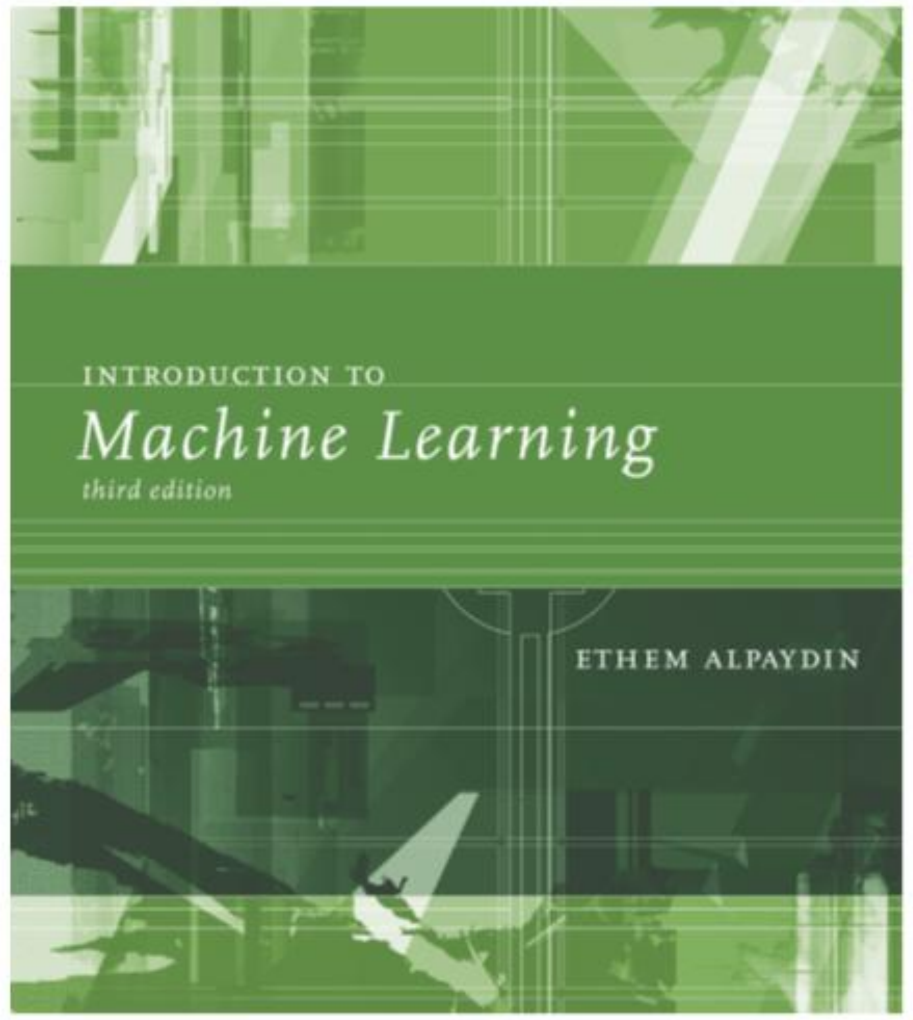
- To learn the basic machine learning techniques, both from a theoretical and practical perspective
- To practice implementing and using these techniques for simple problems
- To understand the advantages/disadvantages of machine learning algorithms and how they relate to each other
- To get an idea on machine learning researches and potential applications

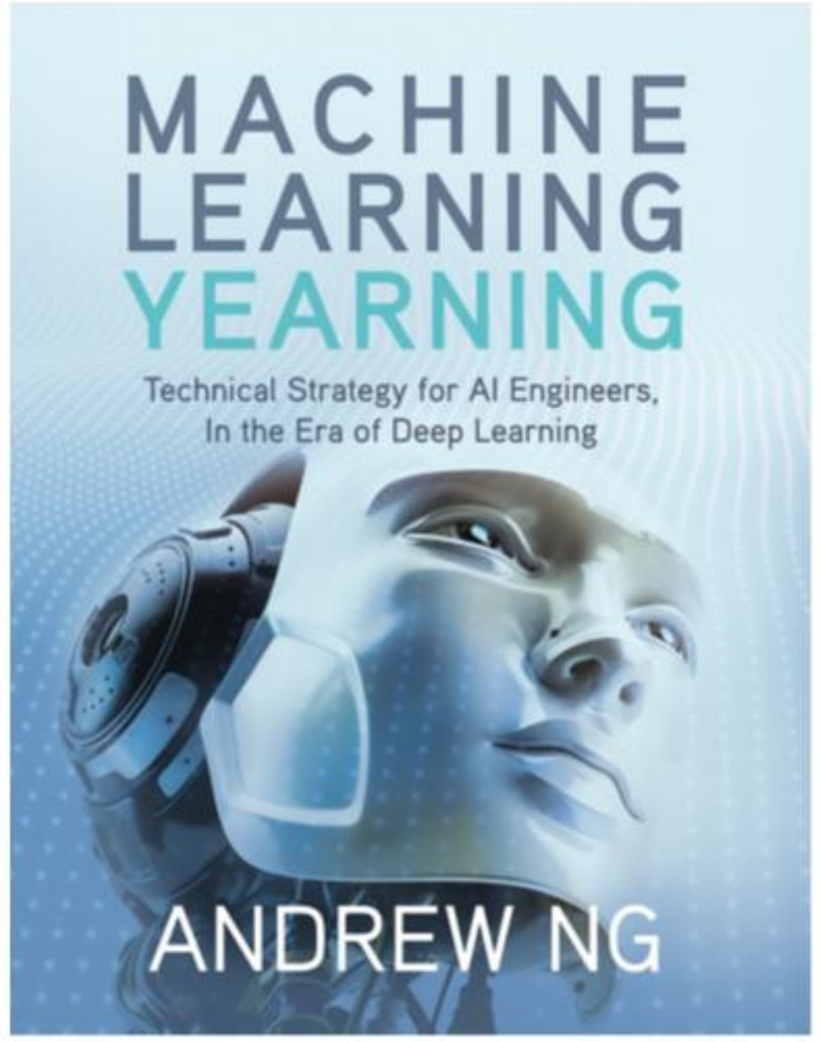
Grading

Class Attendance	: 30
Assignments/Quizzes	: 45
Mid-semester Exam	: 75
Semester-final Exam	: 150
Total	: 300

Books







1. Pattern Recognition and Machine Learning, Christopher M. Bishop
2. Introduction to Machine Learning, Third Edition, Ethem Alpaydin
3. Foundations of Machine Learning, second edition, Mehryar Mohri, Afshin Rostamizadeh, Ameet Talwalkar
4. Machine Learning Yearning, Online version, 2018 by Andrew Ng

Machine Learning Research

Conferences:

– International Conference on Machine Learning (ICML)
– ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)
– Conference on Neural Information Processing Systems (NIPS)
– Asian Conference on Machine Learning (ACML)
– European Conference on Machine Learning (ECML)
– CVPR, ICCV, ECCV, ACCV, WACV, BMVC, MICCAI

Journals:

– IEEE Transactions on Knowledge and Data Engineering (TKDE)
– Journal of Machine Learning Research (JMLR)
– Journal of Artificial Intelligence Research (JAIR)
– IEEE Transactions on Medical Imaging(TMI)
– IEEE Transactions on Pattern Analysis and Machine Intelligence(TPAMI)

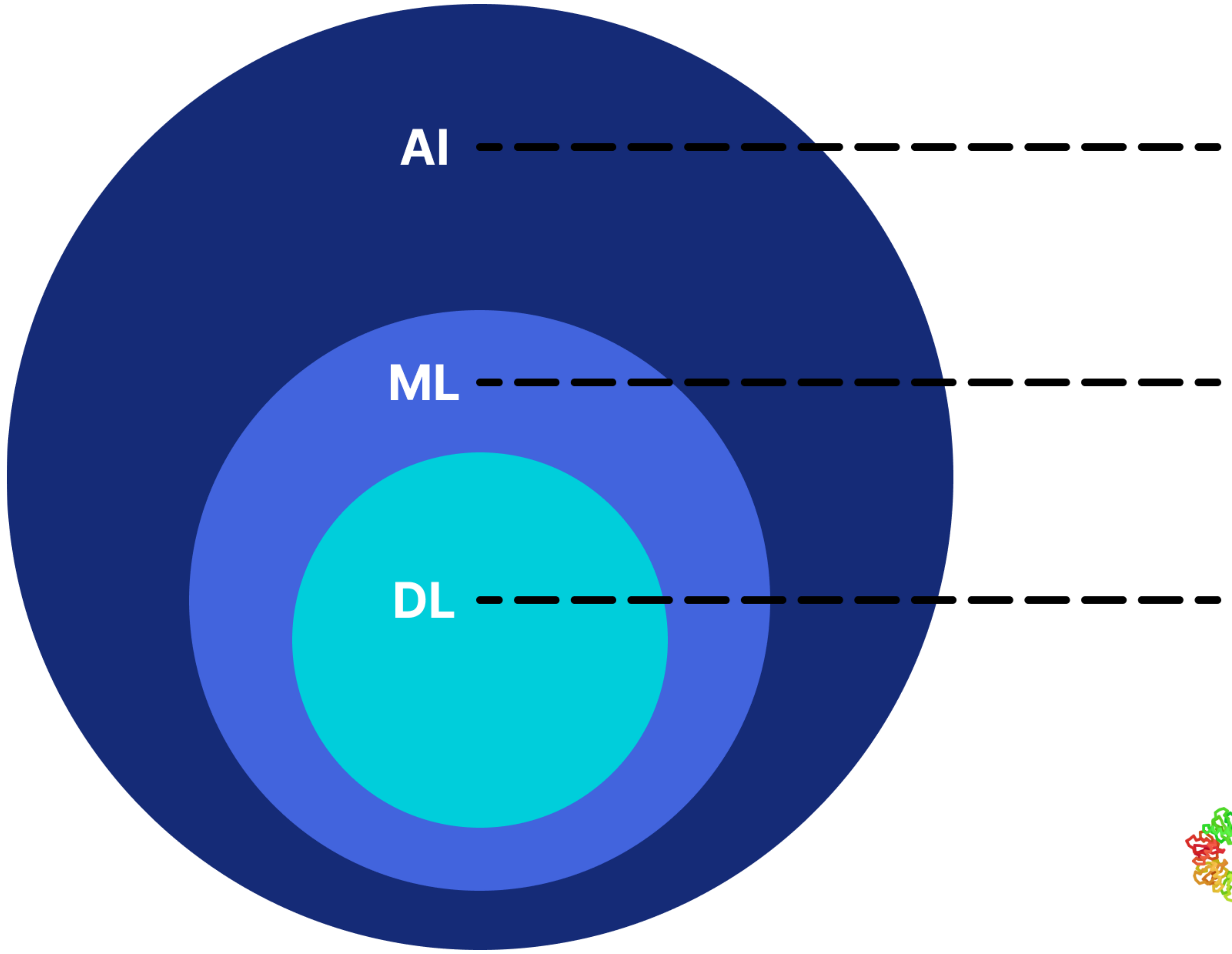
Introduction

Introduction

Machine Learning

Artificial Intelligence

Deep Learning



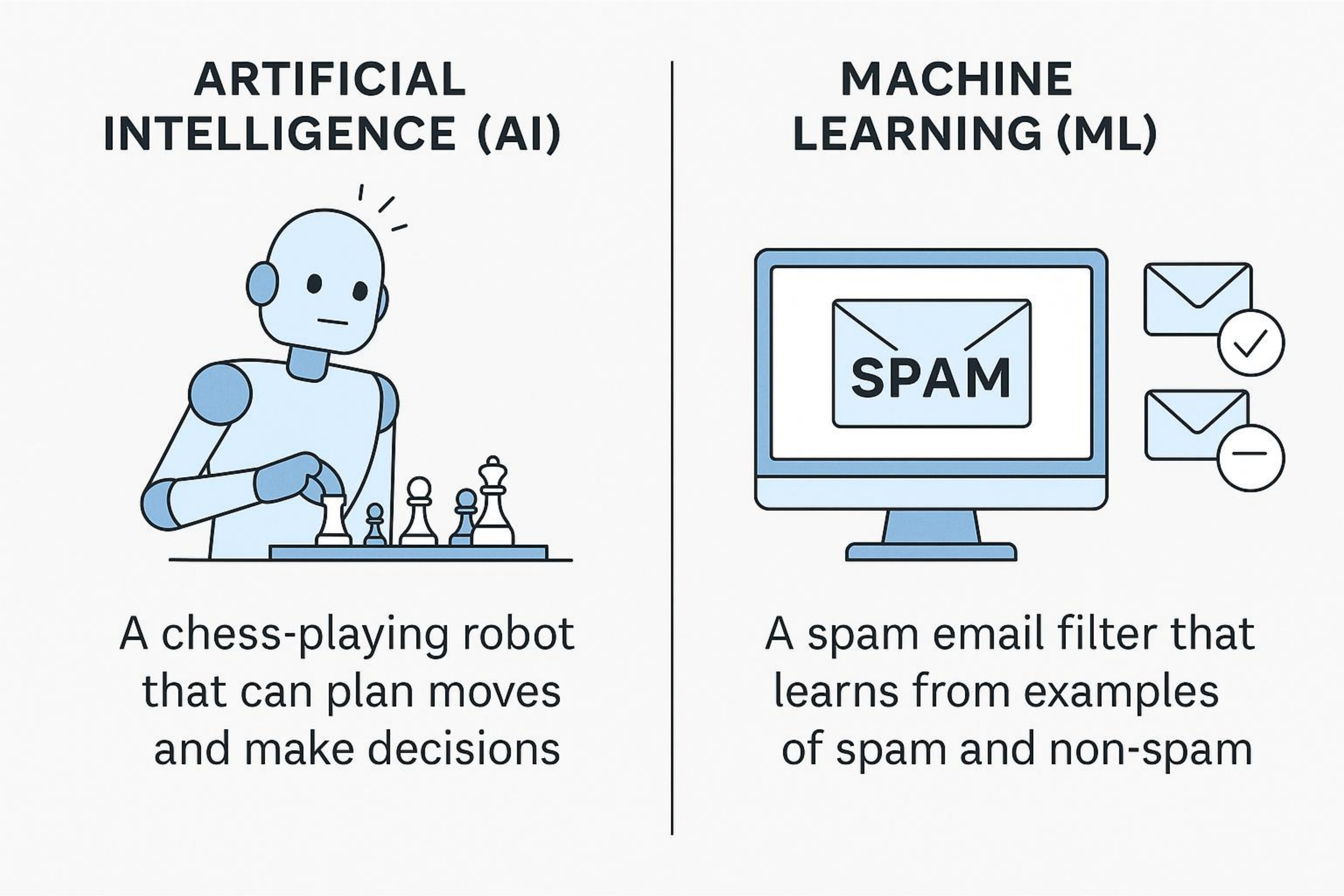
Artificial Intelligence
Engineering of making intelligent machines and programs.

Machine Learning
Ability to learn without being explicitly programmed.

Deep Learning
Learning based on deep neural networks.

neurosnap.ai

Introduction



What is Machine Learning

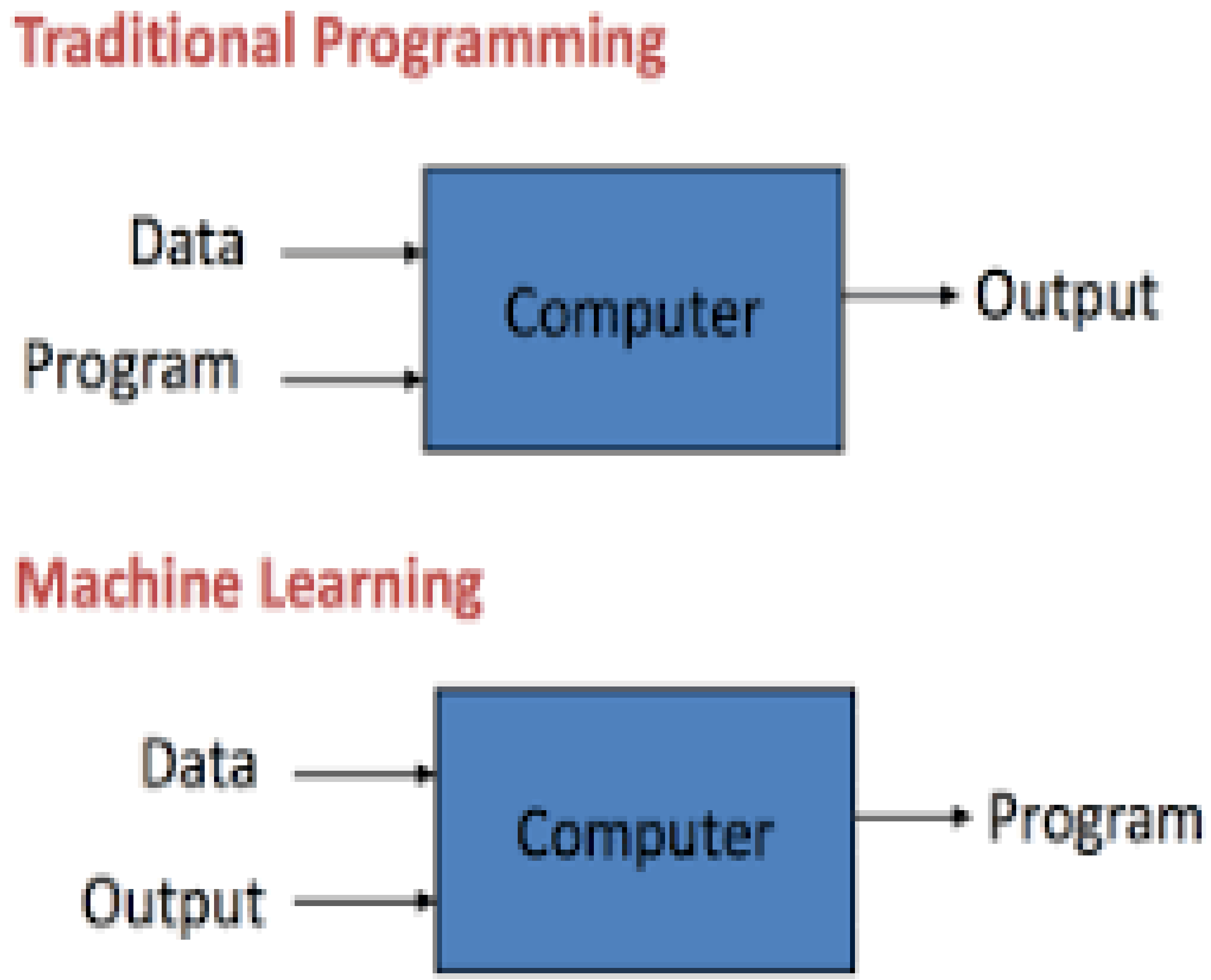
What is Machine Learning

- **Arthur Samuel (1959)**, a pioneer in the field of artificial intelligence and computer gaming, coined the term **"Machine Learning"** as – **"Field of study that gives computers the capability to learn without being explicitly programmed"**.

How it is different from traditional Programming:

- In Traditional Programming, we feed the Input, Program logic and run the program to get output.

- In Machine Learning, we feed the input, output and run it on machine during training and the machine creates its own logic, which is being evaluated while testing.



What is Machine Learning

- **Tom Mitchell (1998)**: A computer program is said to learn from **experience E** with respect to some **task T** and some **performance measure P**, if its **performance on T**, as **measured by P**, **improves** with **experience E**.

Machine Learning is the study of algorithms that
- improve their **performance P**
- at some **task T**
- with **experience E**

Examples

Example 1 **Learning to recognize faces**

– T: recognize faces
– P: % of correct recognitions
– E: opportunity to makes guesses and being told what the truth is

Example 2 **Learning to find clusters in data**

– T: finding clusters
– P: compactness of groups detected
– E: analyses of a growing set of data

Examples

Example 3 **A handwriting recognition learning problem**

- T: recognizing and classifying handwritten words within images
- P: percent of words correctly classified
- E: a database of handwritten words with given classifications

Example 4 **A robot driving learning problem**

- T: driving on public four-lane highways using vision sensors
- P: average distance traveled before an error
- E: a sequence of images and steering commands recorded while observing a human driver

Examples

Example 5 **Suppose your email program watches which emails you do or do not mark as spam and based on that learns how to better filter spam. What is the task T in this setting?**

1. Classifying emails as spam or not spam.
2. Watching you label emails as spam or not spam
3. The number (or fraction) of emails correctly classified as spam/not spam
4. None of the above—this is not a machine learning problem

Why Machine Learning

A large number of tasks cannot be programmed explicitly by humans:

Definitions and Terminology

- **Model:** a specific representation learned from data by applying some machine learning algorithm. A model is also called hypothesis.

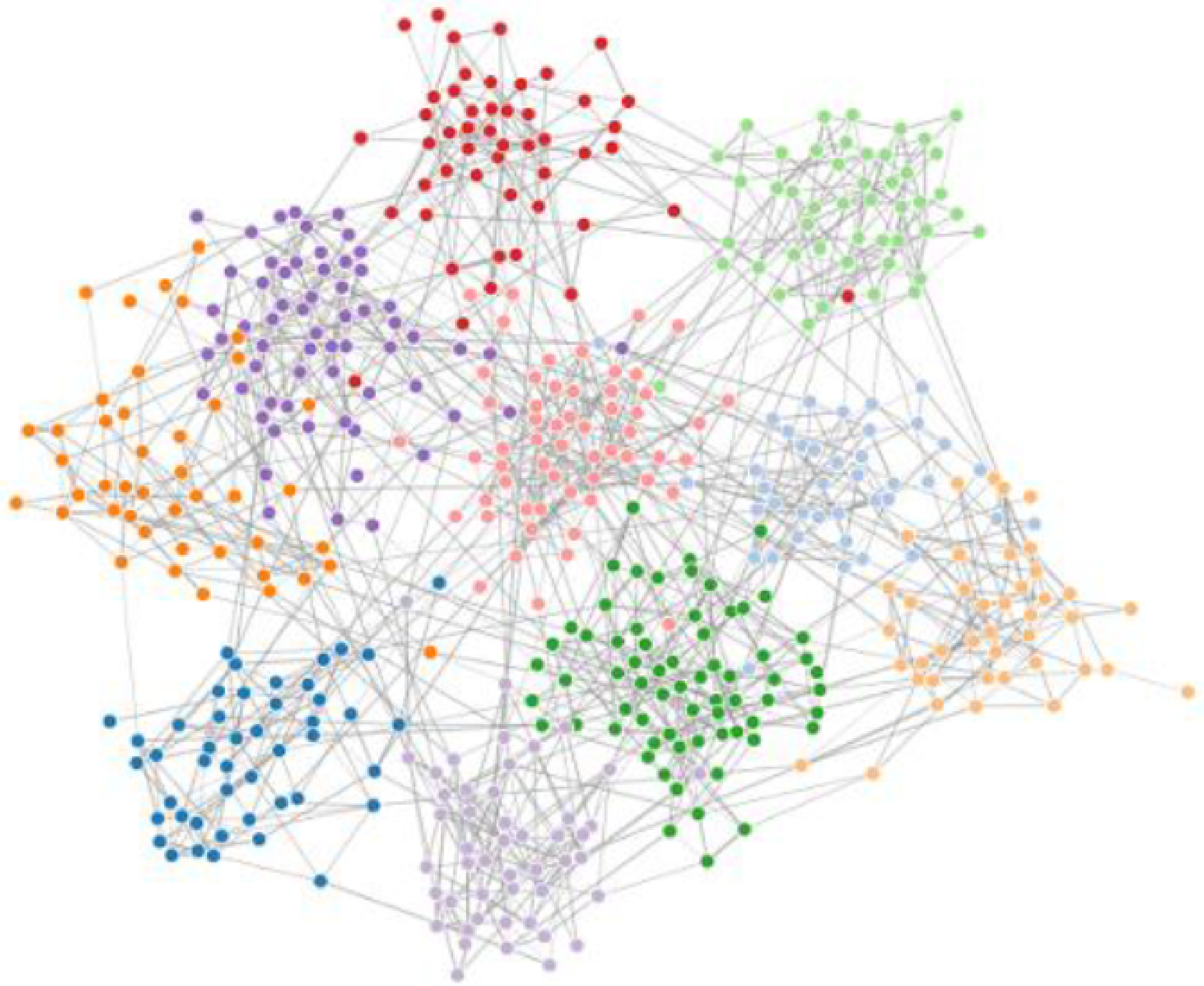
- **Features:** attributes associated to an item, often represented as a vector (e.g., word counts).

- **Labels:** category (classification) or real value (regression) associated to an item.

- **Data:**
  - training data (typically labeled).
  - test data (labeled but labels not seen).
  - validation data (labeled, for tuning parameters).

Basic Steps to Machine Learning

Data



Given to us

Algorithm

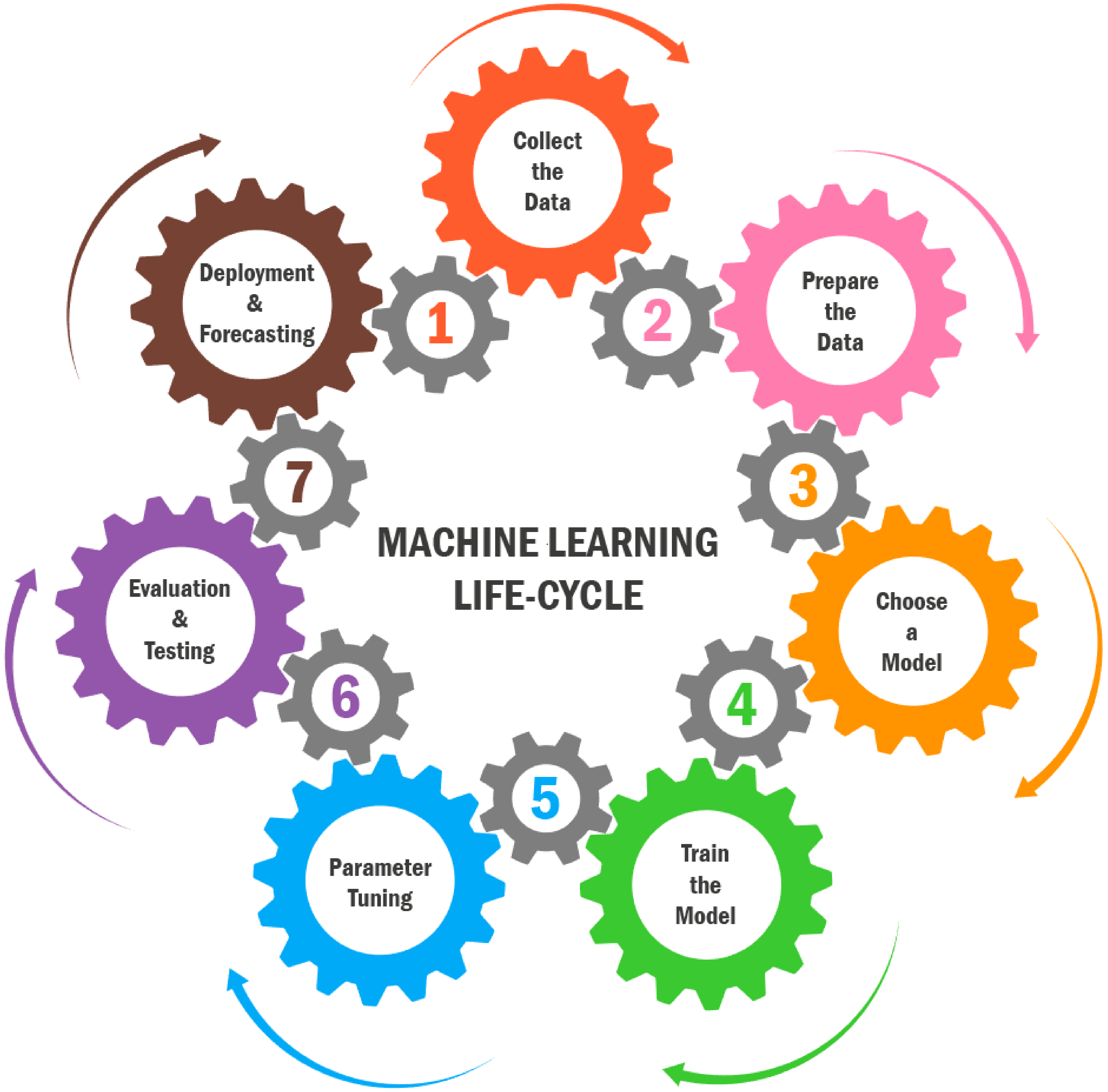
We need to design it

Model

$$f(\mathbf{x})$$

Final output which enables us to make predictions

Machine Learning Life Cycle



Thank You

Any questions?

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